Stage 1 Baseline Results (M/G/k)

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Input Data

```
# -----------------------------
# Input data
# -----------------------------

# Women
prob_women_class2    <- 0.05
total_arrival_women  <- 3.2
arrival_women_class1 <- (1 - prob_women_class2) * total_arrival_women
arrival_women_class2 <- prob_women_class2 * total_arrival_women
service_sigma_women_class1 <- 1.2
service_sigma_women_class2 <- 3.2
service_mean_women_class1  <- 1.67
service_mean_women_class2  <- 5.2

# Men
prob_men_class2    <- 0.05
total_arrival_men  <- 3.2
arrival_men_class1 <- (1 - prob_men_class2) * total_arrival_men
```

```
arrival_men_class2 <- prob_men_class2 * total_arrival_men
service_sigma_men_class1 <- 0.7
service_sigma_men_class2 <- 3.2
service_mean_men_class1  <- 0.83
service_mean_men_class2  <- 5.2

# User preferences
women_pref_GS      <- 0.70
women_with_no_pref <- 0.21
women_pref_unisex  <- 0.09
men_pref_GS        <- 0.70
men_with_no_pref   <- 0.21
men_pref_unisex    <- 0.09

# Routing assumption: 50-50 split for no-preference users
perc_women_no_pref_go_to_GS <- 0.5
perc_men_no_pref_go_to_GS   <- 0.5
```

Functions

```
# ----------------------------
# Functions
# ----------------------------
WAST_computation_general <- function(n, sigma, AST) {
  total_N <- sum(n)
  p_all <- sum(n * AST) / total_N
  variance_terms <- n * (sigma^2 + (AST - p_all)^2)
  sigma_all <- sqrt(sum(variance_terms) / total_N)
  list(mean = p_all, sigma = sigma_all)
}

calculate_ggk_metrics <- function(arrival_rate, service_rate, k, cv_arrival, cv_service) {
  rho <- arrival_rate / (k * service_rate)

  if (rho >= 1) {
    return("System is unstable (Utilization >= 100%)")
  }

  E_s      <- 1 / service_rate
  prob_wait <- rho^(sqrt(2 * (k + 1)) - 1)
```

```r
  ca_sq <- cv_arrival^2
  cs_sq <- cv_service^2

  Wq <- ((ca_sq + cs_sq) / 2) * (prob_wait / (k * (1 - rho))) * E_s
  W  <- Wq + E_s
  Lq <- arrival_rate * Wq
  L  <- arrival_rate * W

  list(
    `Utilization (rho)`             = sprintf("%.2f%%", rho * 100),
    `Pr{Wait > 0}`                  = sprintf("%.6f",   prob_wait),
    `Avg Customers in Queue (Lq)`  = round(Lq, 6),
    `Avg Customers in System (L)`  = round(L,  4),
    `Avg Wait Time in Queue (Wq)`  = round(Wq, 8),
    `Avg Total Time in System (W)` = round(W,  4)
  )
}

print_stats <- function(title, stats) {
  cat("-----------------------------\n")
  cat(title, "\n")
  if (is.character(stats)) {
    cat(stats, "\n")
  } else {
    for (nm in names(stats)) {
      cat(nm, ":", stats[[nm]], "\n")
    }
  }
}

score_table <- data.frame(
  wait = c(
    rep(0, 12), rep(5, 12), rep(10, 12), rep(20, 12), rep(40, 12)
  ),
  gender = c(
    "W","W","W","W","W","W","M","M","M","M","M","M",
    "W","W","W","W","W","W","M","M","M","M","M","M",
    "W","W","W","W","W","W","M","M","M","M","M","M",
    "W","W","W","W","W","W","M","M","M","M","M","M",
    "W","W","W","W","W","W","M","M","M","M","M","M"
  ),
  pref = c(
```

```r
    "GS","GS","NP","NP","UX","UX","GS","GS","NP","NP","UX","UX",
    "GS","GS","NP","NP","UX","UX","GS","GS","NP","NP","UX","UX",
    "GS","GS","NP","NP","UX","UX","GS","GS","NP","NP","UX","UX",
    "GS","GS","NP","NP","UX","UX","GS","GS","NP","NP","UX","UX",
    "GS","GS","NP","NP","UX","UX","GS","GS","NP","NP","UX","UX"
  ),
  uses = c(
    "GS","UX","GS","UX","GS","UX","GS","UX","GS","UX","GS","UX",
    "GS","UX","GS","UX","GS","UX","GS","UX","GS","UX","GS","UX",
    "GS","UX","GS","UX","GS","UX","GS","UX","GS","UX","GS","UX",
    "GS","UX","GS","UX","GS","UX","GS","UX","GS","UX","GS","UX",
    "GS","UX","GS","UX","GS","UX","GS","UX","GS","UX","GS","UX"
  ),
  score = c(
    5,4,5,5,3,5,5,4,5,5,3,5,
    4,3,4,4,2,4,4,3,4,4,2,4,
    3,2,2,2,1,3,3,2,2,2,1,3,
    1,1,1,1,0,2,2,0,1,1,0,2,
    0,0,0,0,0,1,0,0,0,0,0,1
  ),
  stringsAsFactors = FALSE
)

get_score <- function(wait, gender, pref, uses) {
  valid_waits <- c(0, 5, 10, 20, 40)
  lower_waits <- valid_waits[valid_waits <= wait]
  actual_wait <- if (length(lower_waits) == 0) 0 else max(lower_waits)

  row <- score_table[
    score_table$wait   == actual_wait &
    score_table$gender == gender      &
    score_table$pref   == pref        &
    score_table$uses   == uses,
  ]

  if (nrow(row) == 0) return(0)
  row$score[1]
}
```

Config 1 — Status Quo (20W + 10 Men Urinals + 10 Men Stalls)

Config 1 is the status quo with no unisex stalls. All users are routed to gender-segregated restrooms. Women's stalls serve both Class 1 and Class 2, requiring WAST pooling. Men's urinals and stalls serve single classes only.

```
# ------------------------------
# Config 1 - WAST
# ------------------------------

# Women's stalls: pool Class 1 and Class 2
res_women_c1 <- WAST_computation_general(
  n     = c(arrival_women_class1, arrival_women_class2),
  sigma = c(service_sigma_women_class1, service_sigma_women_class2),
  AST   = c(service_mean_women_class1,  service_mean_women_class2)
)
service_mean_all_women_c1  <- res_women_c1$mean
service_sigma_all_women_c1 <- res_women_c1$sigma
cv_women_c1 <- service_sigma_all_women_c1 / service_mean_all_women_c1

# Men's urinals: Class 1 only - no pooling needed
cv_men_class1 <- service_sigma_men_class1 / service_mean_men_class1

# Men's stalls: Class 2 only - no pooling needed
cv_men_class2 <- service_sigma_men_class2 / service_mean_men_class2

cat("Config 1 - WAST\n")
```

```
Config 1 - WAST
```

```
cat("Women stalls : mean =", round(service_mean_all_women_c1, 4),
    " sigma =", round(service_sigma_all_women_c1, 4),
    " CV =", round(cv_women_c1, 4), "\n")
```

```
Women stalls : mean = 1.8465  sigma = 1.5722  CV = 0.8515
```

```
cat("Men urinals  : mean =", service_mean_men_class1,
    " sigma =", service_sigma_men_class1,
    " CV =", round(cv_men_class1, 4), "\n")
```

```
Men urinals  : mean = 0.83  sigma = 0.7  CV = 0.8434
```

```r
cat("Men stalls   : mean =", service_mean_men_class2,
    " sigma =", service_sigma_men_class2,
    " CV =", round(cv_men_class2, 4), "\n")
```

```
Men stalls   : mean = 5.2  sigma = 3.2  CV = 0.6154
```

```r
# -----------------------------
# Config 1 - Queuing Metrics
# -----------------------------

# Women's stalls (k = 20)
womens_stats_c1 <- calculate_ggk_metrics(
  arrival_rate = total_arrival_women,
  service_rate = 1 / service_mean_all_women_c1,
  k            = 20,
  cv_arrival   = 1.0,
  cv_service   = cv_women_c1
)

# Men's urinals (k = 10, Class 1 only)
mens_class1_stats_c1 <- calculate_ggk_metrics(
  arrival_rate = arrival_men_class1,
  service_rate = 1 / service_mean_men_class1,
  k            = 10,
  cv_arrival   = 1.0,
  cv_service   = cv_men_class1
)

# Men's stalls (k = 10, Class 2 only)
mens_class2_stats_c1 <- calculate_ggk_metrics(
  arrival_rate = arrival_men_class2,
  service_rate = 1 / service_mean_men_class2,
  k            = 10,
  cv_arrival   = 1.0,
  cv_service   = cv_men_class2
)

cat("Config 1 - Queuing Metrics\n")
```

```
Config 1 - Queuing Metrics
```

```r
print_stats("Women's stalls   (k=20)", womens_stats_c1)
```

```
------------------------------
Women's stalls   (k=20)
Utilization (rho) : 29.54%
Pr{Wait > 0} : 0.001253
Avg Customers in Queue (Lq) : 0.000453
Avg Customers in System (L) : 5.9093
Avg Wait Time in Queue (Wq) : 0.00014156
Avg Total Time in System (W) : 1.8466
```

```r
print_stats("Men's urinals    (k=10)", mens_class1_stats_c1)
```

```
------------------------------
Men's urinals    (k=10)
Utilization (rho) : 25.23%
Pr{Wait > 0} : 0.006208
Avg Customers in Queue (Lq) : 0.001793
Avg Customers in System (L) : 2.525
Avg Wait Time in Queue (Wq) : 0.00058967
Avg Total Time in System (W) : 0.8306
```

```r
print_stats("Men's stalls     (k=10)", mens_class2_stats_c1)
```

```
------------------------------
Men's stalls     (k=10)
Utilization (rho) : 8.32%
Pr{Wait > 0} : 0.000103
Avg Customers in Queue (Lq) : 6e-06
Avg Customers in System (L) : 0.832
Avg Wait Time in Queue (Wq) : 4.046e-05
Avg Total Time in System (W) : 5.2
```

```r
# ------------------------------
# Config 1 - Fairness Scores
# All users routed to GS (no unisex available)
# ------------------------------
wq_women_c1      <- womens_stats_c1$`Avg Wait Time in Queue (Wq)`
wq_men_class1_c1 <- mens_class1_stats_c1$`Avg Wait Time in Queue (Wq)`
```

```r
wq_men_class2_c1 <- mens_class2_stats_c1$`Avg Wait Time in Queue (Wq)`

average_totalitarian_c1 <- (
  total_arrival_women * women_pref_GS      * get_score(wq_women_c1,      "W", "GS", "GS") +
  total_arrival_women * women_with_no_pref * get_score(wq_women_c1,      "W", "NP", "GS") +
  total_arrival_women * women_pref_unisex  * get_score(wq_women_c1,      "W", "UX", "GS") +
  arrival_men_class1  * men_pref_GS        * get_score(wq_men_class1_c1, "M", "GS", "GS") +
  arrival_men_class1  * men_with_no_pref   * get_score(wq_men_class1_c1, "M", "NP", "GS") +
  arrival_men_class1  * men_pref_unisex    * get_score(wq_men_class1_c1, "M", "UX", "GS") +
  arrival_men_class2  * men_pref_GS        * get_score(wq_men_class2_c1, "M", "GS", "GS") +
  arrival_men_class2  * men_with_no_pref   * get_score(wq_men_class2_c1, "M", "NP", "GS") +
  arrival_men_class2  * men_pref_unisex    * get_score(wq_men_class2_c1, "M", "UX", "GS")
) / (total_arrival_women + total_arrival_men)

all_scores_c1 <- c(
  get_score(wq_women_c1,      "W", "GS", "GS"),
  get_score(wq_women_c1,      "W", "NP", "GS"),
  get_score(wq_women_c1,      "W", "UX", "GS"),
  get_score(wq_men_class1_c1, "M", "GS", "GS"),
  get_score(wq_men_class1_c1, "M", "NP", "GS"),
  get_score(wq_men_class1_c1, "M", "UX", "GS"),
  get_score(wq_men_class2_c1, "M", "GS", "GS"),
  get_score(wq_men_class2_c1, "M", "NP", "GS"),
  get_score(wq_men_class2_c1, "M", "UX", "GS")
)
rawlsian_c1 <- min(all_scores_c1)

cat("Config 1 - Fairness Scores\n")
```

```
Config 1 - Fairness Scores
```

```r
cat(sprintf("Totalitarian (Tr) = %.4f\n", average_totalitarian_c1))
```

```
Totalitarian (Tr) = 4.8200
```

```r
cat(sprintf("Rawlsian     (Mr) = %.4f\n", rawlsian_c1))
```

```
Rawlsian     (Mr) = 3.0000
```

```
cat("Individual scores:", all_scores_c1, "\n")
```

```
Individual scores: 5 5 3 5 5 3 5 5 3
```

```
cat("Worst-off group score:", rawlsian_c1,
    "- UX-preference users forced into GS\n")
```

```
Worst-off group score: 3 - UX-preference users forced into GS
```

Config 2 — All Unisex (40 UX)

Config 2 converts all 40 fixtures to unisex stalls. All four streams (W-C1, W-C2, M-C1, M-C2) share one pooled queue. GS-preference users are into unisex.

```
# -----------------------------
# Config 2 - WAST
# Pool all 4 streams into one
# -----------------------------
sample_sizes_c2 <- c(
  arrival_women_class1,
  arrival_women_class2,
  arrival_men_class1,
  arrival_men_class2
)
sigmas_c2 <- c(
  service_sigma_women_class1,
  service_sigma_women_class2,
  service_sigma_men_class1,
  service_sigma_men_class2
)
ASTs_c2 <- c(
  service_mean_women_class1,
  service_mean_women_class2,
  service_mean_men_class1,
  service_mean_men_class2
)

res_unisex_c2 <- WAST_computation_general(sample_sizes_c2, sigmas_c2, ASTs_c2)
service_mean_unisex_c2  <- res_unisex_c2$mean
service_sigma_unisex_c2 <- res_unisex_c2$sigma
```

```r
cv_unisex_c2 <- service_sigma_unisex_c2 / service_mean_unisex_c2

cat("Config 2 - WAST (all 4 streams pooled)\n")
```

```
Config 2 - WAST (all 4 streams pooled)
```

```r
cat("Unisex stalls : mean =", round(service_mean_unisex_c2, 4),
    " sigma =", round(service_sigma_unisex_c2, 4),
    " CV =", round(cv_unisex_c2, 4), "\n")
```

```
Unisex stalls : mean = 1.4475  sigma = 1.5289  CV = 1.0562
```

```r
cat("Note: CV > 1 - pooling fast (men C1, AST=0.83) and slow (C2, AST=5.2)",
    "creates wider spread than exponential\n")
```

```
Note: CV > 1 - pooling fast (men C1, AST=0.83) and slow (C2, AST=5.2) creates wider spread tl
```

```r
# -----------------------------
# Config 2 - Queuing Metrics
# One line only (k = 40)
# -----------------------------
unisex_stats_c2 <- calculate_ggk_metrics(
  arrival_rate = sum(sample_sizes_c2),
  service_rate = 1 / service_mean_unisex_c2,
  k            = 40,
  cv_arrival   = 1.0,
  cv_service   = cv_unisex_c2
)

cat("Config 2 - Queuing Metrics\n")
```

```
Config 2 - Queuing Metrics
```

```r
print_stats("Unisex stalls (k=40)", unisex_stats_c2)
```

```
-----------------------------
Unisex stalls (k=40)
Utilization (rho) : 23.16%
```

```
Pr{Wait > 0} : 0.000008
Avg Customers in Queue (Lq) : 2e-06
Avg Customers in System (L) : 9.264
Avg Wait Time in Queue (Wq) : 3.8e-07
Avg Total Time in System (W) : 1.4475
```

```r
# ----------------------------
# Config 2 - Fairness Scores
# All users go to UX (only option)
# ----------------------------
wq_unisex_c2 <- unisex_stats_c2$`Avg Wait Time in Queue (Wq)`

average_totalitarian_c2 <- (
  total_arrival_women * women_pref_GS      * get_score(wq_unisex_c2, "W", "GS", "UX") +
  total_arrival_women * women_with_no_pref * get_score(wq_unisex_c2, "W", "NP", "UX") +
  total_arrival_women * women_pref_unisex  * get_score(wq_unisex_c2, "W", "UX", "UX") +
  arrival_men_class1  * men_pref_GS        * get_score(wq_unisex_c2, "M", "GS", "UX") +
  arrival_men_class1  * men_with_no_pref   * get_score(wq_unisex_c2, "M", "NP", "UX") +
  arrival_men_class1  * men_pref_unisex    * get_score(wq_unisex_c2, "M", "UX", "UX") +
  arrival_men_class2  * men_pref_GS        * get_score(wq_unisex_c2, "M", "GS", "UX") +
  arrival_men_class2  * men_with_no_pref   * get_score(wq_unisex_c2, "M", "NP", "UX") +
  arrival_men_class2  * men_pref_unisex    * get_score(wq_unisex_c2, "M", "UX", "UX")
) / (total_arrival_women + total_arrival_men)

all_scores_c2 <- c(
  get_score(wq_unisex_c2, "W", "GS", "UX"),
  get_score(wq_unisex_c2, "W", "NP", "UX"),
  get_score(wq_unisex_c2, "W", "UX", "UX"),
  get_score(wq_unisex_c2, "M", "GS", "UX"),
  get_score(wq_unisex_c2, "M", "NP", "UX"),
  get_score(wq_unisex_c2, "M", "UX", "UX")
)
rawlsian_c2 <- min(all_scores_c2)

cat("Config 2 - Fairness Scores\n")
```

```
Config 2 - Fairness Scores
```

```r
cat(sprintf("Totalitarian (Tr) = %.4f\n", average_totalitarian_c2))
```

```
Totalitarian (Tr) = 4.3000
```

```
cat(sprintf("Rawlsian     (Mr) = %.4f\n", rawlsian_c2))
```

```
Rawlsian     (Mr) = 4.0000
```

```
cat("Individual scores:", all_scores_c2, "\n")
```

```
Individual scores: 4 5 5 4 5 5
```

```
cat("Worst-off group score:", rawlsian_c2,
    "- GS-preference users (70%) forced into UX\n")
```

```
Worst-off group score: 4 - GS-preference users (70%) forced into UX
```

Config 3 — Hybrid (15W + 10 Men Urinals + 5 Men Stalls + 10 UX)

Config 3 introduces 10 unisex stalls alongside gender-segregated ones. GS route: 80.5% of each gender (70% GS-pref + 50% x 21% no-pref). Unisex route: 19.5% of each gender (9% UX-pref + 50% x 21% no-pref).

```
# -----------------------------
# Config 3 - Arrival Allocation
# -----------------------------
gs_route   <- women_pref_GS + perc_women_no_pref_go_to_GS * women_with_no_pref  # 0.805
ux_route   <- women_pref_unisex + (1 - perc_women_no_pref_go_to_GS) * women_with_no_pref  # (

# Women GS route
arrival_women_class1_gs_c3 <- total_arrival_women * (1 - prob_women_class2) * gs_route
arrival_women_class2_gs_c3 <- total_arrival_women * prob_women_class2 * gs_route

# Men GS route
arrival_men_class1_gs_c3   <- total_arrival_men * (1 - prob_men_class2) * gs_route
arrival_men_class2_gs_c3   <- total_arrival_men * prob_men_class2 * gs_route

# Unisex route (all 4 streams)
arrival_women_class1_ux_c3 <- total_arrival_women * (1 - prob_women_class2) * ux_route
arrival_women_class2_ux_c3 <- total_arrival_women * prob_women_class2 * ux_route
arrival_men_class1_ux_c3   <- total_arrival_men * (1 - prob_men_class2) * ux_route
arrival_men_class2_ux_c3   <- total_arrival_men * prob_men_class2 * ux_route

cat("Config 3 - Arrival Allocation\n")
```

Config 3 - Arrival Allocation

```r
cat("GS route proportion  :", gs_route, "\n")
```

```
GS route proportion  : 0.805
```

```r
cat("UX route proportion  :", ux_route, "\n")
```

```
UX route proportion  : 0.195
```

```r
cat("Women GS total lambda:", arrival_women_class1_gs_c3 + arrival_women_class2_gs_c3, "\n")
```

```
Women GS total lambda: 2.576
```

```r
cat("Men GS total lambda  :", arrival_men_class1_gs_c3   + arrival_men_class2_gs_c3,   "\n")
```

```
Men GS total lambda  : 2.576
```

```r
cat("Unisex total lambda  :", arrival_women_class1_ux_c3 + arrival_women_class2_ux_c3 +
                              arrival_men_class1_ux_c3   + arrival_men_class2_ux_c3,   "\n")
```

```
Unisex total lambda  : 1.248
```

```r
# -----------------------------
# Config 3 - WAST
# Women's stalls: same pool as Config 1 (ratio preserved)
# Unisex stalls:  same pool as Config 2 (ratio preserved)
# Men's urinals and stalls: single class - no pooling
# -----------------------------

# Women's stalls
res_women_c3 <- WAST_computation_general(
  n     = c(arrival_women_class1_gs_c3, arrival_women_class2_gs_c3),
  sigma = c(service_sigma_women_class1,  service_sigma_women_class2),
  AST   = c(service_mean_women_class1,   service_mean_women_class2)
)
service_mean_all_women_c3  <- res_women_c3$mean
service_sigma_all_women_c3 <- res_women_c3$sigma
```

```r
cv_women_c3 <- service_sigma_all_women_c3 / service_mean_all_women_c3

# Unisex stalls: pool all 4 streams
sample_sizes_ux_c3 <- c(
  arrival_women_class1_ux_c3, arrival_women_class2_ux_c3,
  arrival_men_class1_ux_c3,   arrival_men_class2_ux_c3
)
sigmas_ux_c3 <- c(
  service_sigma_women_class1, service_sigma_women_class2,
  service_sigma_men_class1,   service_sigma_men_class2
)
ASTs_ux_c3 <- c(
  service_mean_women_class1, service_mean_women_class2,
  service_mean_men_class1,   service_mean_men_class2
)
res_unisex_c3 <- WAST_computation_general(sample_sizes_ux_c3, sigmas_ux_c3, ASTs_ux_c3)
service_mean_unisex_c3  <- res_unisex_c3$mean
service_sigma_unisex_c3 <- res_unisex_c3$sigma
cv_unisex_c3 <- service_sigma_unisex_c3 / service_mean_unisex_c3

cat("Config 3 - WAST\n")
```

```
Config 3 - WAST
```

```r
cat("Women's stalls : mean =", round(service_mean_all_women_c3, 4),
    " CV =", round(cv_women_c3, 4), "\n")
```

```
Women's stalls : mean = 1.8465  CV = 0.8515
```

```r
cat("Men's urinals  : mean =", service_mean_men_class1,
    " CV =", round(cv_men_class1, 4), "\n")
```

```
Men's urinals  : mean = 0.83  CV = 0.8434
```

```r
cat("Men's stalls   : mean =", service_mean_men_class2,
    " CV =", round(cv_men_class2, 4), "\n")
```

```
Men's stalls   : mean = 5.2  CV = 0.6154
```

```r
cat("Unisex stalls  : mean =", round(service_mean_unisex_c3, 4),
    " CV =", round(cv_unisex_c3, 4), "\n")
```

```
Unisex stalls  : mean = 1.4475  CV = 1.0562
```

```r
# -----------------------------
# Config 3 - Queuing Metrics
# -----------------------------

# Women's stalls (k = 15)
womens_stats_c3 <- calculate_ggk_metrics(
  arrival_rate = arrival_women_class1_gs_c3 + arrival_women_class2_gs_c3,
  service_rate = 1 / service_mean_all_women_c3,
  k            = 15,
  cv_arrival   = 1.0,
  cv_service   = cv_women_c3
)

# Men's urinals (k = 10, Class 1 GS route)
mens_class1_stats_c3 <- calculate_ggk_metrics(
  arrival_rate = arrival_men_class1_gs_c3,
  service_rate = 1 / service_mean_men_class1,
  k            = 10,
  cv_arrival   = 1.0,
  cv_service   = cv_men_class1
)

# Men's stalls (k = 5, Class 2 GS route)
mens_class2_stats_c3 <- calculate_ggk_metrics(
  arrival_rate = arrival_men_class2_gs_c3,
  service_rate = 1 / service_mean_men_class2,
  k            = 5,
  cv_arrival   = 1.0,
  cv_service   = cv_men_class2
)

# Unisex stalls (k = 10)
unisex_stats_c3 <- calculate_ggk_metrics(
  arrival_rate = sum(sample_sizes_ux_c3),
  service_rate = 1 / service_mean_unisex_c3,
  k            = 10,
```

```r
  cv_arrival   = 1.0,
  cv_service   = cv_unisex_c3
)

cat("Config 3 - Queuing Metrics\n")
```

```
Config 3 - Queuing Metrics
```

```r
print_stats("Women's stalls (k=15)", womens_stats_c3)
```

```
-----------------------------
Women's stalls (k=15)
Utilization (rho) : 31.71%
Pr{Wait > 0} : 0.004755
Avg Customers in Queue (Lq) : 0.001905
Avg Customers in System (L) : 4.7585
Avg Wait Time in Queue (Wq) : 0.00073933
Avg Total Time in System (W) : 1.8472
```

```r
print_stats("Men's urinals  (k=10)", mens_class1_stats_c3)
```

```
-----------------------------
Men's urinals  (k=10)
Utilization (rho) : 20.31%
Pr{Wait > 0} : 0.002788
Avg Customers in Queue (Lq) : 0.000608
Avg Customers in System (L) : 2.0318
Avg Wait Time in Queue (Wq) : 0.00024847
Avg Total Time in System (W) : 0.8302
```

```r
print_stats("Men's stalls   (k=5)",  mens_class2_stats_c3)
```

```
-----------------------------
Men's stalls   (k=5)
Utilization (rho) : 13.40%
Pr{Wait > 0} : 0.007059
Avg Customers in Queue (Lq) : 0.000753
Avg Customers in System (L) : 0.6705
Avg Wait Time in Queue (Wq) : 0.00584312
Avg Total Time in System (W) : 5.2058
```

```r
print_stats("Unisex stalls  (k=10)", unisex_stats_c3)
```

```
-----------------------------
Unisex stalls  (k=10)
Utilization (rho) : 18.06%
Pr{Wait > 0} : 0.001809
Avg Customers in Queue (Lq) : 0.000422
Avg Customers in System (L) : 1.8069
Avg Wait Time in Queue (Wq) : 0.00033803
Avg Total Time in System (W) : 1.4478
```

```r
# -----------------------------
# Config 3 - Fairness Scores
# GS-pref to GS, UX-pref to UX
# No-pref: 50% to GS, 50% to UX
# -----------------------------
wq_women_gs_c3  <- womens_stats_c3$`Avg Wait Time in Queue (Wq)`
wq_men_ur_c3    <- mens_class1_stats_c3$`Avg Wait Time in Queue (Wq)`
wq_men_st_c3    <- mens_class2_stats_c3$`Avg Wait Time in Queue (Wq)`
wq_unisex_c3    <- unisex_stats_c3$`Avg Wait Time in Queue (Wq)`

average_totalitarian_c3 <- (
  # Women GS route
  total_arrival_women * women_pref_GS      * get_score(wq_women_gs_c3, "W", "GS", "GS") +
  total_arrival_women * women_with_no_pref * perc_women_no_pref_go_to_GS *
                        get_score(wq_women_gs_c3, "W", "NP", "GS") +
  # Women UX route
  total_arrival_women * women_with_no_pref * (1 - perc_women_no_pref_go_to_GS) *
                        get_score(wq_unisex_c3, "W", "NP", "UX") +
  total_arrival_women * women_pref_unisex  * get_score(wq_unisex_c3, "W", "UX", "UX") +
  # Men GS route (urinals - Class 1)
  arrival_men_class1  * men_pref_GS        * get_score(wq_men_ur_c3, "M", "GS", "GS") +
  arrival_men_class1  * men_with_no_pref   * perc_men_no_pref_go_to_GS *
                        get_score(wq_men_ur_c3, "M", "NP", "GS") +
  # Men GS route (stalls - Class 2)
  arrival_men_class2  * men_pref_GS        * get_score(wq_men_st_c3, "M", "GS", "GS") +
  arrival_men_class2  * men_with_no_pref   * perc_men_no_pref_go_to_GS *
                        get_score(wq_men_st_c3, "M", "NP", "GS") +
  # Men UX route (Class 1)
  arrival_men_class1  * men_with_no_pref   * (1 - perc_men_no_pref_go_to_GS) *
                        get_score(wq_unisex_c3, "M", "NP", "UX") +
```

```r
  arrival_men_class1  * men_pref_unisex    * get_score(wq_unisex_c3, "M", "UX", "UX") +
  # Men UX route (Class 2)
  arrival_men_class2  * men_with_no_pref   * (1 - perc_men_no_pref_go_to_GS) *
                        get_score(wq_unisex_c3, "M", "NP", "UX") +
  arrival_men_class2  * men_pref_unisex    * get_score(wq_unisex_c3, "M", "UX", "UX")
) / (total_arrival_women + total_arrival_men)

all_scores_c3 <- c(
  get_score(wq_women_gs_c3, "W", "GS", "GS"),
  get_score(wq_women_gs_c3, "W", "NP", "GS"),
  get_score(wq_unisex_c3,   "W", "NP", "UX"),
  get_score(wq_unisex_c3,   "W", "UX", "UX"),
  get_score(wq_men_ur_c3,   "M", "GS", "GS"),
  get_score(wq_men_ur_c3,   "M", "NP", "GS"),
  get_score(wq_men_st_c3,   "M", "GS", "GS"),
  get_score(wq_men_st_c3,   "M", "NP", "GS"),
  get_score(wq_unisex_c3,   "M", "NP", "UX"),
  get_score(wq_unisex_c3,   "M", "UX", "UX")
)
rawlsian_c3 <- min(all_scores_c3)

cat("Config 3 - Fairness Scores\n")
```

```
Config 3 - Fairness Scores
```

```r
cat(sprintf("Totalitarian (Tr) = %.4f\n", average_totalitarian_c3))
```

```
Totalitarian (Tr) = 5.0000
```

```r
cat(sprintf("Rawlsian     (Mr) = %.4f\n", rawlsian_c3))
```

```
Rawlsian     (Mr) = 5.0000
```

```r
cat("Individual scores:", all_scores_c3, "\n")
```

```
Individual scores: 5 5 5 5 5 5 5 5 5 5
```

Cross-Configuration Comparison

```
# ------------------------------
# Cross-configuration summary
# ------------------------------
cat("=== Cross-Configuration Comparison ===\n\n")
```

```
=== Cross-Configuration Comparison ===
```

```
cat(sprintf("%-30s  Config 1  Config 2  Config 3\n", "Metric"))
```

```
Metric                          Config 1  Config 2  Config 3
```

```
cat(sprintf("%-30s  --------  --------  --------\n", ""))
```

```
                                --------  --------  --------
```

```
cat(sprintf("%-30s  %8.4f  %8.4f  %8.4f\n",
    "Totalitarian (Tr)",
    average_totalitarian_c1,
    average_totalitarian_c2,
    average_totalitarian_c3))
```

```
Totalitarian (Tr)                 4.8200    4.3000    5.0000
```

```
cat(sprintf("%-30s  %8.4f  %8.4f  %8.4f\n",
    "Rawlsian (Mr)",
    rawlsian_c1,
    rawlsian_c2,
    rawlsian_c3))
```

```
Rawlsian (Mr)                     3.0000    4.0000    5.0000
```

```
cat("\n")
```

```r
cat("Totalitarian ranking: Config 3 > Config 1 > Config 2\n")
```

```
Totalitarian ranking: Config 3 > Config 1 > Config 2
```

```r
cat("Rawlsian ranking:     Config 3 > Config 2 > Config 1\n")
```

```
Rawlsian ranking:     Config 3 > Config 2 > Config 1
```

```r
cat("Both frameworks recommend Config 3 as the most just configuration.\n")
```

```
Both frameworks recommend Config 3 as the most just configuration.
```